

ANSWER KEY

GATE CS — GATE CS 2026 Trend-Based Mock Paper
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Q.No.	Type	Answer	Marks
1	3MCQ	B	3
2	2MCQ	A	2
3	2MCQ	A	2
4	2MCQ	C	2
5	3MCQ	C	3
6	2MSQ	30	2
7	2MCQ	A	2
8	2MCQ	24	2
9	2MSQ	B	2
10	2MCQ	B	2
11	2NAT	C	2
12	2NAT	B	2
13	2MSQ	C	2
14	1MCQ	B	1
15	1MCQ	B	1
16	1NAT	B	1
17	1MCQ	C	1
18	1MCQ	A	1
19	1NAT	C	1
20	1NAT	C	1
21	1MCQ	C	1
22	1MCQ	B	1
23	1MCQ	B	1
24	2MSQ	C	2
25	1NAT	B	1
26	1NAT	A	1
27	1MCQ	B	1
28	2MCQ	C	2
29	1MCQ	B	1
30	1MCQ	A	1
31	1MCQ	C	1
32	1NAT	B	1
33	1MCQ	C	1
34	1MCQ	B	1
35	1MCQ	B	1
36	1MCQ	C	1

37	1MCQ	B	1
38	1MCQ	B	1
39	1MCQ	B	1
40	1NAT	B	1
41	1MCQ	A	1
42	1MCQ	B	1
43	1NAT	B	1
44	1NAT	B	1
45	1MCQ	C	1
46	1MCQ	A	1
47	2NAT	64 Mbps	2
48	2NAT	$a^n b^n$	2
49	2MSQ	A	2
50	2MSQ	435	2
51	2MSQ	B	2
52	2MCQ	30.06 seconds	2
53	2MCQ	C	2
54	2MCQ	B	2
55	2MCQ	C	2
56	2MCQ	B	2
57	2MCQ	B	2
58	2NAT	There are three RAW data hazards (I1!'I2, I2!'I3, I3!'I4)	2
59	1NAT	A	1
60	2NAT	Optimal (Belady's optimal) page replacement policy	2
61	2NAT	6	2
62	2NAT	0	2
63	2MCQ	A	2
64	2MCQ	C	2
65	2MCQ	0	2

Q1. General Aptitude | 3 marks | 3MCQ

Let 1 and 2 denote two arbitrary prime numbers. Which one of the following statements is correct for all values of 1 and...

Answer: B

For any prime p , p^2 is composite, so statement B is always true; others have counter examples.

Source: Adapted from GATE 2025 CS Session 2 Q5

Q2. General Aptitude | 2 marks | 2MCQ

The paper as shown in the figure is folded to make a cube where each square corresponds to a particular face of the cube...

Answer: A

The arrangement of faces in option A matches the adjacency shown in the net.

Source: Adapted from GATE 2025 CS Session 2 Q4

Q3. General Aptitude | 2 marks | 2MCQ

Consider the relationships among P, Q, R, S, and T: $\ddot{y} P$ is the brother of Q. $\ddot{y} S$ is the daughter of Q. $\ddot{y} T$ is the sister..

Answer: A

R is Q's mother, making her S's grandmother; P is Q's brother, thus uncle of S and T. Statements 3 and 4 are false.

Source: Adapted from GATE 2025 CS Session 1 Q4

Q4. General Aptitude | 2 marks | 2MCQ

Despite his initial hesitation, Rehman's _____ to contribute to the success of the project never wavered. Select the...

Answer: C

'Resolve' fits the context of unwavering determination.

Source: Adapted from GATE 2025 CS Session 2 Q1

Q5. General Aptitude | 3 marks | 3MCQ

According to the map shown in the figure, which one of the following statements is correct? Note: The figure shown is re...

Answer: C

Based on the relative positions shown in the map, the chemistry lab lies to the southeast of the physics lab, making statement C the only correct one.

Source: Adapted from GATE 2025 CS Session 1 Q5

Q6. General Aptitude | 2 marks | 2MSQ

The average marks obtained by a class in an examination were calculated as 30.8. However, while checking the marks enter...

Answer: 30

Let n be the number of students. Original total = $30.8 / n$. After correcting the 24 to 42, total increases by 18, giving $(30.8 / n + 18) / n = 31.4$. Solving $30.8 / n + 18 = 31.4 / n \Rightarrow 18 = 0.6 / n \Rightarrow n = 30$.

Source: Adapted from GATE 2025 CS Session 1 Q3

Q7. General Aptitude | 2 marks | 2MCQ

Ravi had _____ younger brother who taught at _____ university. He was widely regarded as _____ honorable man. Select ...

Answer: A

The sentence requires indefinite articles: "a younger brother" (first blank), "a university" (second blank), and because "honorable" begins with a vowel sound, "an honorable man" (third blank). Hence the correct sequence is a; a; an, which corresponds to option A.

Source: Adapted from GATE 2025 CS Session 1 Q1

Q8. General Aptitude | 2 marks | 2MCQ
Two wizards try to create a spell using all the four elements, water, air, fire, and earth. For this, they decide to mix...

Answer: 24

With four distinct elements, the number of possible orders is the number of permutations: $4! = 24$. Each wizard independently tries every permutation, so each makes 24 attempts.

Source: Adapted from GATE 2024 CS Session 2 Q2

Q9. General Aptitude | 2 marks | 2MSQ
Synonym of 'ubiquitous':

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q10

Q10. Engineering Mathematics | 2 marks | 2MCQ
Derivative of x^3 at $x=2$:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q1

Q11. Engineering Mathematics | 2 marks | 2NAT
Let A be a 3×3 matrix with eigenvalues 1, 2, and 3. Then the trace of A^2 is:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q1

Q12. Engineering Mathematics | 2 marks | 2NAT
Mean of binomial $n=10, p=0.5$:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q1

Q13. Engineering Mathematics | 2 marks | 2MSQ
Probability of at least one head in 3 coin tosses:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q11

Q14. Engineering Mathematics | 1 marks | 1MCQ
Determinant of 2×2 identity:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q1

Q15. Programming and Data Structures | 1 marks | 1MCQ
Infix to postfix uses:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q6

Q16. Programming and Data Structures | 1 marks | 1NAT
What is the time complexity of inserting at the beginning of a linked list?

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q3

Q17. Programming and Data Structures | 1 marks | 1MCQ
Avg hash lookup:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2021 CS Session 1 Q3

Q18. Programming and Data Structures | 1 marks | 1MCQ
Inorder traversal of BST gives:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q3

Q19. Algorithms | 1 marks | 1NAT
LCS uses:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2023 CS Session 1 Q7

Q20. Algorithms | 1 marks | 1NAT
Merge sort space:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2021 CS Session 1 Q7

Q21. Algorithms | 1 marks | 1MCQ
Kruskal uses:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2022 CS Session 1 Q2

Q22. Algorithms | 1 marks | 1MCQ
Time complexity of binary search:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q11

Q23. Algorithms | 1 marks | 1MCQ
BFS uses:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q7

Q24. Algorithms | 2 marks | 2MSQ
The time complexity of Dijkstra's algorithm using a binary heap is:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q7

Q25. Algorithms | 1 marks | 1NAT
Worst-case time complexity of merge sort:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q12

Q26. Operating Systems | 1 marks | 1NAT
Round-robin uses:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q4

Q27. Operating Systems | 1 marks | 1MCQ
Banker's algorithm is for:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q4

Q28. Operating Systems | 2 marks | 2MCQ
Using 3 frames, which page replacement algorithm would not cause any page fault for reference string 0,1,2,3,0,1,4,0,1,2...

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q8

Q29. Operating Systems | 1 marks | 1MCQ
Inode contains:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2022 CS Session 1 Q4

Q30. Operating Systems | 1 marks | 1MCQ
TLB stands for:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2022 CS Session 1 Q9

Q31. Operating Systems | 1 marks | 1MCQ
Which of the following is NOT a necessary condition for deadlock?

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q4

Q32. Operating Systems | 1 marks | 1NAT
Semaphore with value 1 is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q8

Q33. Computer Networks | 1 marks | 1MCQ
Asymmetric encryption:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2022 CS Session 1 Q5

Q34. Computer Networks | 1 marks | 1MCQ
CSMA/CD is used in:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q13

Q35. Computer Networks | 1 marks | 1MCQ
Protocol to find MAC from IP:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q6

Q36. Computer Networks | 1 marks | 1MCQ
Which protocol operates at the application layer of the TCP/IP model?

Answer: C

Correct Answer: C

Source: Adapted from GATE 2024 CS Session 1 Q6

Q37. Computer Networks | 1 marks | 1MCQ
TCP connection establishment uses:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q14

Q38. Computer Networks | 1 marks | 1MCQ
OSPF is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q6

Q39. Digital Logic | 1 marks | 1MCQ
The number of Boolean functions of n variables is:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q2

Q40. Digital Logic | 1 marks | 1NAT
Full adder has how many inputs:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q2

Q41. Digital Logic | 1 marks | 1MCQ
Flip-flop stores:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2021 CS Session 1 Q2

Q42. Theory of Computation | 1 marks | 1MCQ
Halting problem:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2021 CS Session 1 Q8

Q43. Theory of Computation | 1 marks | 1NAT
PDA recognizes:

Answer: B

Correct Answer: B

Source: Adapted from GATE 2023 CS Session 1 Q14

Q44. Theory of Computation | 1 marks | 1NAT
Which language is regular?

Answer: B

Correct Answer: B

Source: Adapted from GATE 2024 CS Session 1 Q13

Q45. Compiler Design | 1 marks | 1MCQ
3-address code operands:

Answer: C

Correct Answer: C

Source: Adapted from GATE 2021 CS Session 1 Q9

Q46. Compiler Design | 1 marks | 1MCQ
Lexical analyzer outputs:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q15

Q47. CS | 2 marks | 2NAT
8 bytes in 1 cycle from a device to main memory through cycle stealing at regular intervals. Which one of the following ...

Answer: 64 Mbps

If the processor runs at 100 /MHz, 1% of its cycles are 1 /MHz. Each DMA cycle transfers 8 /bytes = 64 /bits, so the DMA rate = 1 /MHz $\times$ 64 /bits = 64 / $\times$ /10⁶ /bits/s = 64 /Mbps.

Source: Adapted from GATE 2024 CS Session 2 Q11

Q48. CS | 2 marks | 2NAT

Consider the pushdown automaton (PDA) P below, which runs on the input alphabet $\{a, b\}$, has stack alphabet $\{A\}$, and ...

Answer: $a^n b^n$

Each 'a' pushes an A onto the stack (first a pushes the initial A, subsequent a's duplicate the top A, effectively adding one A per a). Each 'b' pops an A. The stack is emptied exactly when the number of b's equals the number of a's, so the PDA accepts the language $\{a^n b^n \mid n \geq 0\}$.

Source: Adapted from GATE 2023 CS Session 1 Q40

Q49. CS | 2 marks | 2MSQ

Let f be a 3-variable Boolean function that produces output as '1' when at least two of the input variables are '1'. Which...

Answer: A

The function is the 3 variable majority (at least two 1s). It is associative, so $M(a,b,M(c,d,e)) = M(M(a,b,c),d,e)$ holds for all Boolean inputs. The other identities fail on counter examples.

Source: Adapted from GATE 2025 CS Session 1 Q24

Q50. CS | 2 marks | 2MSQ

The pseudocode of a function $\text{fun}()$ is given below: $\text{fun}(\text{int } A[0, \dots, n-1])$ { for $i=0$ to $n-2$ for $j=0$ to $n-i-2$ if $(A[i] > A[j+1])$...

Answer: 435

The algorithm is bubble sort; it swaps adjacent elements for each inversion. A descending array of 30 distinct elements has $30 \times 29 / 2 = 435$ inversions, so 435 swaps occur.

Source: Adapted from GATE 2025 CS Session 1 Q33

Q51. CS | 2 marks | 2MSQ

Consider a Boolean expression given by $(x, y, z) = (3, 5, 6, 7)$. Which of the following statements is/are CORRECT?

Answer: B

The minterms 3 (011), 5 (101), 6 (110) and 7 (111) are exactly the cases where at least two of the three inputs are 1. Simplifying $\sum m(3, 5, 6, 7)$ gives the Boolean expression $f(a,b,c) = ab + bc + ca$. Hence option B is correct. Option A states equality with the complement $\sum m(0, 1, 2, 4) = \overline{f}$, which is false, and options C and D claim the function is independent of the inputs, which is also false.

Source: Adapted from GATE 2024 CS Session 1 Q47

Q52. CS | 2 marks | 2MCQ

Consider a disk with the following specifications: rotation speed of 6000 RPM, average seek time of 5 milliseconds, 500 ...

Answer: 30.06 seconds

Each random sector requires average seek (5 /ms) + average rotational latency (half of 10 /ms = 5 /ms) + transfer time (10 /ms per revolution / 500 sectors = 0.02 /ms). Total per sector = 10.02 /ms = 0.01002 /s. For 3000 sectors: $3000 \times 0.01002 /s = 30.06 /s$, rounded to two decimal places.

Source: Adapted from GATE 2024 CS Session 2 Q53

Q53. CS | 2 marks | 2MCQ

A square paper, shown in figure (I), is folded along the dotted lines as shown in the figures (II) and (III). Then a few...

Answer: C

Folding the square along the dotted lines creates four congruent quadrants. The cuts made in the folded state are mirrored in each quadrant when the paper is unfolded, producing a pattern that is symmetric about both the horizontal and vertical axes. Among the given options, only pattern C exhibits this four fold symmetry, matching the expected result.

Source: Adapted from GATE 2025 CS Session 1 Q9

Q54. CS | 2 marks | 2MCQ

Let $R1$ and $R2$ be two 4-bit registers that store numbers in 2's complement form. For the operation $R1 + R2$, which one of the...

Answer: B

Overflow occurs when adding two numbers with the same sign yields a result with opposite sign. In option B, $R1 = 1100$ (-4) and $R2 = 1010$ (-6) sum to -10, which cannot be represented in 4-bit two's complement (range -8 to +7). The binary addition gives a positive result (0110) indicating overflow.

Source: Adapted from GATE 2022 CS Session 1 Q18

Q55. CS | 2 marks | 2MCQ

Consider a digital display system (DDS) shown in the figure that displays the contents of register X. A 16-bit code word...

Answer: C

S has 1024 words, requiring a $10 \text{ to } 2^{10}$ ($10 \text{ to } 1024$) decoder; R has 32 words, requiring a $5 \text{ to } 2^5$ ($5 \text{ to } 32$) decoder. The mode bit M selects between the two sources, which is performed by a $2 \text{ to } 1$ multiplexer. Hence P is a $10 \text{ to } 1024$ decoder, Q is a $5 \text{ to } 32$ decoder, and T is a $2 \text{ to } 1$ multiplexer.

Source: Adapted from GATE 2022 CS Session 1 Q40

Q56. General Aptitude | 2 marks | 2MCQ

Let 1 and 2 denote two arbitrary prime numbers. Which one of the following statements is correct for all values of 1 and...

Answer: B

Since 1 and 2 represent arbitrary prime numbers p and q , the expression "12" denotes their product $p \cdot q$. The product of two primes (both greater than 1) is always composite, never prime. Hence the statement "12 is not a prime number" holds for all choices of p and q , making option B the correct choice.

Source: Adapted from GATE 2025 CS Session 2 Q5

Q57. CS | 2 marks | 2MCQ

$= \{0, 1, 2, 3, \dots\}$ is the set of non-negative integers. Let be the set of functions from to itself. For any two functi...

Answer: B

The set of all functions N^N is closed under pointwise addition, which is associative and has the zero function as identity. However, most functions lack additive inverses within N (only the zero function is invertible), so the structure is not a group. Since addition of natural numbers is commutative, the operation is Abelian. Hence $(F, +)$ forms an Abelian monoid only.

Source: Adapted from GATE 2025 CS Session 1 Q49

Q58. CS | 2 marks | 2NAT

An instruction format has the following structure: Instruction Number: Opcode destination reg, source reg-1, source reg...

Answer: There are three RAW data hazards (I1!'I2, I2!'I3, I3!'I4)

I2 reads R3 written by I1, I3 reads R5 written by I2, and I4 reads R3 written by I3, creating three read after write dependencies.

Source: Adapted from GATE 2024 CS Session 2 Q31

Q59. Programming and Data Structures | 1 marks | 1NAT

Inorder traversal of BST gives:

Answer: A

Correct Answer: A

Source: Adapted from GATE 2023 CS Session 1 Q3

Q60. CS | 2 marks | 2NAT

Consider a demand paging system with three frames, and the following page reference string: 1 2 3 4 5 4 1 6 4 5 1 3 2. T...

Answer: Optimal (Belady's optimal) page replacement policy

The fault pattern (page faults at references 1,2,3,4,5,8,11,12,13) matches the decisions of the optimal algorithm: replace the page whose next use is farthest in the future. Simulating optimal replacement on the given reference string yields exactly the same sequence of page replacements, while FIFO, LRU, Clock, etc., produce different fault patterns.

Source: Adapted from GATE 2025 CS Session 2 Q47

Q61. CS | 2 marks | 2NAT

Let $= \{1,2,3,4\}$. For a , let $a()$ be the product of symbols in modulo 7. We take $1() = 1$, where ϵ is the null string. For examp...

Answer: 6

The DFA needs to track the product modulo 7, which can be any of the six non-zero residues $\{1,2,3,4,5,6\}$. Each residue is distinguishable because a suitable continuation can make the product equal 2 for one residue and not for another. Hence the minimal DFA has 6 states.

Source: Adapted from GATE 2025 CS Session 2 Q60

Q62. CS | 2 marks | 2NAT

The maximum value of such that the edge between the nodes B and C is included in every minimum spanning tree of the given...

Answer: 0

The question does not provide the specific graph or edge weights, so the maximum value cannot be determined from the given information. Without the graph details, we cannot compute the integer required.

Source: Adapted from GATE 2025 CS Session 1 Q64

Q63. CS | 2 marks | 2MCQ

In the context of owner and weak entity sets in the ER (Entity-Relationship) data model, which one of the following statements...

Answer: A

In ER modeling, a weak entity cannot exist without being linked to an owner via the identifying relationship, so it must participate totally in that relationship. The owner entity, however, may exist without any associated weak entities, so its participation can be partial. Hence only statement A is true.

Source: Adapted from GATE 2024 CS Session 2 Q20

Q64. CS | 2 marks | 2MCQ

"I put the brown paper in my pocket along with the chalks, and possibly other things. I suppose every one must have referred..."

Answer: C

The passage explicitly describes the pocket-knife as "the infant of the sword," making option C the only statement supported by the text.

Source: Adapted from GATE 2025 CS Session 1 Q6

Q65. CS | 2 marks | 2MCQ

The value of the definite integral $\int_{-3}^3 \int_{-2}^2 \int_{-1}^1 (4x^2y - z^3) dz dy dx$ is . (Rounded off to the nearest integer) CS GATE 202...

Answer: 0

The integral is $\int_{x=-3}^3 \int_{y=-2}^2 \int_{z=-1}^1 (4x^2y - z^3) dz dy dx$. The inner z integral of $4x^2y - z^3$ gives $8x^2y$, which is odd in y and integrates to zero over symmetric limits $[-2, 2]$. The z integral of $-z^3$ also vanishes because z^3 is odd over $[-1, 1]$. Hence the whole triple integral equals 0, which rounds to the nearest integer 0.

Source: Adapted from GATE 2023 CS Session 1 Q31